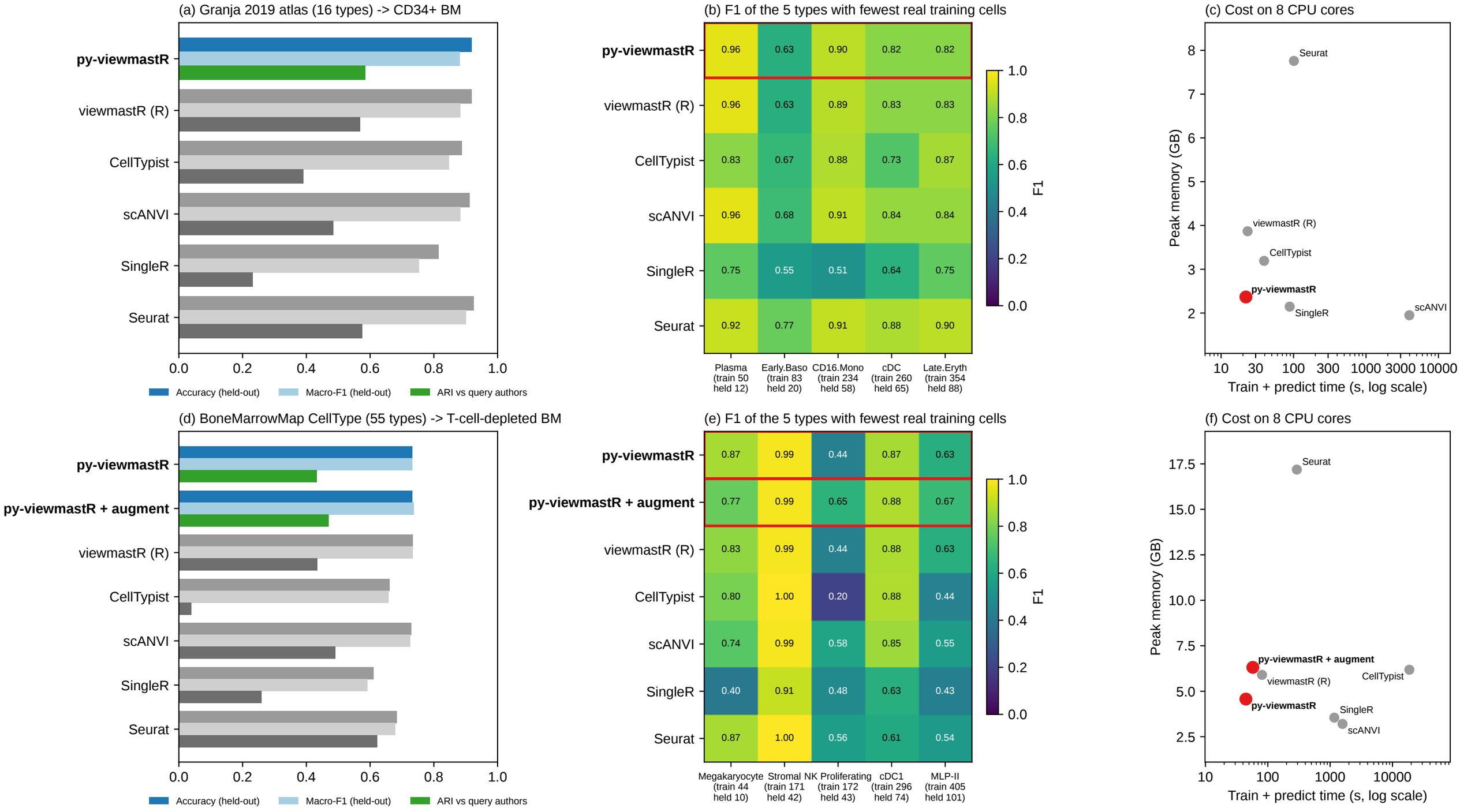
(a) Granja 2019 atlas (16 types) -> CD34+ BM
py-viewmastR
viewmastR (R)
CellTypist
scANVI
SingleR
Seurat
0.0
0.2
0.4
0.6
0.8
1.0
Accuracy (held-out)
Macro-F1 (held-out)
ARI vs query authors
(b) F1 of the 5 types with fewest real training cells
0.96 0.63 0.90 0.82 0.82
0.96 0.63 0.89 0.83 0.83
0.83 0.67 0.88 0.73 0.87
0.96 0.68 0.91 0.84 0.84
0.75 0.55 0.51 0.64 0.75
0.92 0.77 0.91 0.88 0.90
Plasma (train 50 held 12)
Early.Baso (train 83 held 20)
CD16.Mono (train 234 held 58)
cDC (train 260 held 65)
Late.Eryth (train 354 held 88)
F1
(c) Cost on 8 CPU cores
Peak memory (GB)
Train + predict time (s, log scale)
Seurat
viewmastR (R)
CellTypist
py-viewmastR
SingleR
scANVI
(d) BoneMarrowMap CellType (55 types) -> T-cell-depleted BM
py-viewmastR
py-viewmastR + augment
viewmastR (R)
CellTypist
scANVI
SingleR
Seurat
0.0
0.2
0.4
0.6
0.8
1.0
Accuracy (held-out)
Macro-F1 (held-out)
ARI vs query authors
(e) F1 of the 5 types with fewest real training cells
0.87 0.99 0.44 0.87 0.63
0.77 0.99 0.65 0.88 0.67
0.83 0.99 0.44 0.88 0.63
0.80 1.00 0.20 0.88 0.44
0.74 0.99 0.58 0.85 0.55
0.40 0.91 0.48 0.63 0.43
0.87 1.00 0.56 0.61 0.54
Megakaryocyte (train 44 held 10)
Stromal (train 171 held 42)
NK Proliferating (train 172 held 43)
cDC1 (train 296 held 74)
MLP-II (train 405 held 101)
F1
(f) Cost on 8 CPU cores
Peak memory (GB)
Train + predict time (s, log scale)
Seurat
py-viewmastR + augment
viewmastR (R)
CellTypist
py-viewmastR
SingleR
scANVI